

Francesco Vinci

Data & Process Science Researcher

+39-388-344-6145 | francesco.vinci@unipd.it | frvinci.github.io

[in francesco-vinci](#) | [franvinci](#) | [Francesco Vinci](#)



Biography

I am a **data and process science** researcher working at the intersection of data-driven and process-oriented techniques to enable *what-if* analysis, process optimization, and prescriptive analytics. My work applies **machine and deep learning** to process modeling and simulation, event-log and model repair, data augmentation, and decision support, mainly in **business process management** and **healthcare**.

I completed my **Ph.D. in Brain, Mind & Computer Science at the University of Padua** (defended in July 2026), advised by Prof. Massimiliano de Leoni. During my doctorate I spent one year in Aachen, Germany, hosted by Fraunhofer FIT and the Process and Data Science (PADS) group at RWTH Aachen, collaborating with Prof. Wil van der Aalst.

Experience

- **University of Padua** [🌐] October 2025 - Current
Research Fellow Padova (PD), Italy
 - I am currently a Postdoctoral Research Fellow ("Assegnista di Ricerca") at the University of Padua. My research focuses on the discovery and enhancement of process simulation models using explainable machine learning techniques, generative and agentic AI, with applications to the optimization of healthcare processes, particularly in emergency medicine.
- **University of Padua** [🌐] October 2022 - September 2025
Ph.D. Researcher Padova (PD), Italy
 - During my Ph.D., I conducted research in process mining, process simulation, and data-driven modeling, with a strong focus on the discovery, repair, and improvement of simulation models using event data. My work leveraged data science and artificial intelligence techniques for online model adaptation and process optimization, and was applied to real-world domains such as business and healthcare processes.
- **Danieli Automation** [🌐] March 2022 - August 2022
R&D Data Science Intern Buttrio (UD), Italy
 - I was an intern at the Danieli Research Center, where I developed a framework combining supervised and unsupervised machine and deep learning methods for video anomaly detection to identify sticking events in the continuous casting process of steel manufacturing.

Visiting

- **RWTH Aachen University** [🌐] March 2025 - September 2025
Visiting Researcher Aachen, Germany
 - During my Ph.D., I visited the Process and Data Science (PADS) Research Group in Aachen, where I conducted research on the discovery of white-box process simulation models. During this period, I also developed *ProSiT*, a tool for configurable process simulation.
 - Supervised by Prof. Wil van der Aalst.
- **Fraunhofer FIT** [🌐] September 2024 - March 2025
Visiting Researcher Aachen, Germany
 - During my Ph.D., I visited the Fraunhofer Process Mining Group in Aachen, where I conducted research on the discovery of white-box process simulation models and their adaptation in online settings.
 - Supervised by Prof. Wil van der Aalst and Prof. Gyunam Park.

Teaching

• University of Padua

April 2023 - Current

Teaching Assistant

Padova (PD), Italy

- I am a Teaching Assistant for the Database course in the Bachelor's degree in Computer Science, where I support students during laboratory sessions on PostgreSQL and pgAdmin, and contribute to the preparation of class and project materials.

Education

• University of Padua

October 2022 - July 2026

Ph.D. in Brain, Mind & Computer Science

Padova (PD), Italy

- Curriculum: Computer Science for Societal Challenges and Innovation
- Thesis: *Reliable and Explainable Process Simulation Models: Discovery, Refinement, and Applications*
- Supervisor: Prof. Massimiliano de Leoni
- Co-supervisor: Dr. Silvia Gabrielli
- Thesis reviewers: Prof. Marlon Dumas and Prof. Andrea Marrella
- Defense committee: Prof. Davide Bacciu, Prof. Andrea Marrella, Prof. Andrea Burattin
- Defended: cum laude, 16 July 2026

• University of Padua

October 2020 - September 2022

M.Sc. in Data Science

Padova (PD), Italy

- Thesis: *Semi-supervised Deep Learning methods for Video Anomaly Detection applied to sticking identification during steelmaking continuous casting process*
- Supervisor: Prof. Michele Rossi
- Co-supervisor: Andrea Zamolo

• University of Padua

October 2016 - February 2020

B.Sc. in Mathematics

Padova (PD), Italy

- Thesis: *Enlargement of filtrations in discrete time and applications to finance*
- Supervisor: Prof. Claudio Fontana

Publications

C=CONFERENCE, J=JOURNAL, *=UNDER REVIEW

- [J.3*] **Francesco Vinci**, Gyunam Park, Silvia Gabrielli, Wil van der Aalst, Massimiliano de Leoni (2026). **Reliable and Configurable Process Simulation Models: Design, Implementation and User Assessment**. Under Review in *Information Systems* (Elsevier).
- [C.8] **Francesco Vinci**, Massimiliano de Leoni (2026). **Bridging Accuracy and Interpretability in Business Process Simulation via Knowledge Distillation**. In *Business Process Management Forum*. September 27 - October 2, 2026, Toronto, Canada. **Accepted**.
- [J.2] **Francesco Vinci**, Davide Aloini, Elisabetta Benevento, Alessandro Stefanini, Francesca Zen, Massimiliano de Leoni (2026). **Improving Organizational Processes in Healthcare through Simulation-Driven Resource Allocation: Methodology and Real-World Case Study**. In *Artificial Intelligence in Medicine*. Vol. 176. pp. 103391. Elsevier. DOI: 10.1016/j.artmed.2026.103391. Ranking (JCR): Q1.
- [C.7] **Francesco Vinci**, Gyunam Park, Wil van der Aalst, Massimiliano de Leoni (2025). **ProSiT: A Tool for Interactive and Transparent Process Simulations**. In *International Conference on Service Oriented Computing, Demonstrations and Resources*. December 1-4, 2025, Shenzhen, China. **Accepted**.
- [C.6] **Francesco Vinci**, Gyunam Park, Wil van der Aalst, Massimiliano de Leoni (2025). **Reliable and Configurable Process Simulations via Probabilistic White-Box Models**. In *International Conference on Service Oriented Computing*, pp. 337-352. Springer Nature Singapore. December 1-4, 2025, Shenzhen, China. DOI: 10.1007/978-981-95-5015-9_24. Ranking (CORE): A.
- [C.5] Ngoc-Diem Le, Alessandro Padella, **Francesco Vinci**, Massimiliano de Leoni (2025). **Leveraging Counterfactuals for Prescriptive Process Analytics**. In *Business Process Management: Responsible BPM Forum, Process Technology Forum, Educators Forum. BPM 2025*, pp. 200-215. Springer Nature Switzerland. August 31- September 5, 2025, Seville, Spain. DOI: 10.1007/978-3-032-02936-2_15.
- [C.4] **Francesco Vinci**, Gyunam Park, Wil van der Aalst, Massimiliano de Leoni (2025). **Online Discovery of Simulation Models for Evolving Business Processes**. In *International Conference on Business Process Management*, pp. 451-468. Springer Nature Switzerland. August 31- September 5, 2025, Seville, Spain. DOI: 10.1007/978-3-032-02867-9_27. Ranking (CORE): A.

- [J.1] **Francesco Vinci**, Massimiliano de Leoni (2025). **Balancing Fitness and Precision in Process Model Repair: Framework Formalization, Assessment, and Benefits for Simulation**. In *Process Science*, Vol. 2, n. 3, Best Process Science Conference Papers 2024 Collection. Springer Nature. DOI: 10.1007/s44311-025-00007-7.
- [C.3] **Francesco Vinci**, Massimiliano de Leoni (2024). **Repairing Process Models Through Simulation and Explainable AI**. In *International Conference on Business Process Management*, pp. 129-145. Springer Nature Switzerland. September 1-6, 2024, Krakow, Poland. DOI: 10.1007/978-3-031-70396-6_8. Ranking (CORE): A.
- [C.2] Alessandro Padella, Felix Mannhardt, **Francesco Vinci**, Massimiliano de Leoni, Irene Vanderfeesten (2024). **Experience-Based Resource Allocation for Remaining Time Optimization**. In *International Conference on Business Process Management*, pp. 345-362. Springer Nature Switzerland. September 1-6, 2024, Krakow, Poland. DOI: 10.1007/978-3-031-70396-6_20. Ranking (CORE): A.
- [C.1] Massimiliano de Leoni, **Francesco Vinci**, Sander Leemans, Felix Mannhardt (2023). **Investigating the Influence of Data-Aware Process States on Activity Probabilities in Simulation Models: Does Accuracy Improve?**. In *International Conference on Business Process Management*, pp. 129-145. Springer Nature Switzerland. September 11-15, 2023, Utrecht, The Netherlands. DOI: 10.1007/978-3-031-41620-0_8. Ranking (CORE): A.

Tools and Software

- **ProSiT** 2025
Open-Source Process Simulation Tool 
 - Designed and built an open-source web app for data-driven business process simulation with explainable ML, interactive what-if analysis, and discrete-event simulation.
 - Evaluated with 12 BPM domain experts: SUS usability score 81.25/100 ("Good-to-Excellent"); featured as ICSOC 2025 Demo paper.
- **prosit-pm** 2025
Python Library for Process Simulation 
 - Open-source library (pip install prosit-pm) that automatically discovers interpretable simulation parameters from event logs as decision-tree models (arrival rates, execution times, routing probabilities, resource selection).
 - Supports both batch (scikit-learn) and incremental (River) discovery, enabling adaptation to evolving processes.

Conferences & Talks

- **Europe-Korea Conference on Science and Technology (EKC2026)** 22 July 2026
Centre des Congrès Pierre Baudis, Toulouse, France. (Online). 
 - Invited Speaker: *"From Insight to Improvement: Simulation and Optimization of Business Processes"*.
- **23rd International Conference on Service-Oriented Computing (ICSOC 2025)** 1-4 December 2025
Harbin Institute of Technology, Shenzhen, China 
 - Attendee and Speaker
 - Presented the paper *"Reliable and Configurable Process Simulations via Probabilistic White-Box Models"* at the main conference.
 - Presented the paper *"ProSiT: A Tool for Interactive and Transparent Process Simulations"* at the demonstration and resources session.
- **23rd International Conference on Business Process Management (BPM 2025)** August 31 – September 5 2025
Universidad de Sevilla, Seville, Spain 
 - Attendee and Speaker
 - Presented the paper *"Online Discovery of Simulation Models for Evolving Business Processes"* at the main conference.
- **7th International Workshop on Process-Oriented Data Science for Healthcare (PODS4H)** 14 October 2024
Technical University of Denmark, Lyngby, Denmark 
 - Attendee and Speaker
 - Presented *"Healthcare Process Optimization via Simulations: An Emergency Department Case Study"* during the poster session.
- **6th International Conference on Process Mining (ICPM 2024)** 14 October 2024
Technical University of Denmark, Lyngby, Denmark 
 - Attendee
- **22nd International Conference on Business Process Management (BPM 2024)** 1-6 September 2024
AGH University, Krakow, Poland 
 - Attendee and Speaker
 - Presented the paper *"Repairing Process Models Through Simulation and Explainable AI"* at the main conference.

- **12th European Big Data Management & Analytics Summer School (eBISS 2024)** 1-5 July 2024
University of Padua, Padova, Italy 
 ◦ Organizer
- **21st International Conference on Business Process Management (BPM 2023)** 11-15 September 2023
Utrecht University, Utrecht, The Netherlands 
 ◦ Attendee and Speaker
 ◦ Presented the paper "*Investigating the Influence of Data-Aware Process States on Activity Probabilities in Simulation Models: Does Accuracy Improve?*" at the main conference.
- **11th European Big Data Management & Analytics Summer School (eBISS 2023)** 3-7 July 2023
Universitat Politècnica de Catalunya, Barcelona, Spain 
 ◦ Attendee

Hackathons

- **World Data League** July 2021
World Data League 
 ◦ Finalist
- **Youth in Action** April 2021
Fondazione Italiana Accenture 
 ◦ *Premio Promotori* Winner

Skills

- **Programming Languages:** Python, SQL, R, Julia, JavaScript, C, C++
- **ML & AI Frameworks:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CatBoost, LightGBM, SciPy, River (online ML), LangChain, SHAP (explainability)
- **Data Engineering & Analysis:** Pandas, NumPy, event log processing (XES/OCEL), time series analysis, anomaly detection, data augmentation, statistical modeling
- **Process & Simulation:** Discrete-event simulation, process mining (Pm4Py, ProM, Apromore), Petri nets, BPMN, what-if analysis, prescriptive analytics
- **Software & Reproducibility:** Git, Docker, Flask, REST APIs, PostgreSQL, SQLite
- **Visualization & Reporting:** Matplotlib, Seaborn, Plotly, R Studio

Additional Information

Languages: Italian (C2), English (C1)
Driving Licence: B (EEA)